Procedural Verifiability for Legal AI Outputs: An Audit Framework for Case Recommendation

Abstract

Artificial intelligence (AI) systems increasingly determine which legal authorities become visible before legal reasoning begins. This article argues that case recommendation is a procedural-status problem, not a retrieval-accuracy problem. It introduces *normative material screening output*: an output that filters, ranks or summarizes legal authorities without itself becoming a legal source or a decision reason. The framework assigns procedural status through output effect, system-role caps and a six-part audit vector for source and corpus verifiability, query and version traceability, authority hierarchy, ranking and counter-material handling, contestability and adoption logging. The model is implemented in a provider-agnostic audit harness and emits proof-carrying status certificates for each allocation. The evaluation covers 246 scenario packets with 679 embedded items and 148,905 derived robustness checks across public retrieval, raw model outputs, source-supported repairs, adversarial variants, formal invariants, gate ablations, source-chain attacks, contestation challenges, policy replay, metamorphic tests, query diagnostics, ranking counterfactuals, proof certificates, certificate tamper-resistance and coding-uncertainty tests. The central result is a separation: authority coverage and high retrieval recall do not confer procedural status unless the output also preserves a reconstructable source chain, issue-specific authority set, counter-material pathway, contestability route, adoption record and replayable proof object. Across 12 simplified substitute rules, the best source-plus-score-plus-review rule still produced 38 false positives against the protocol-defined reference allocation. The article contributes a formal audit protocol, an executable artifact and a tamper-resistant proof-carrying certificate layer for deciding when legal AI outputs may remain reference information, become professional support, qualify as normative material screening or be downgraded.

Keywords: legal information retrieval; case recommendation; legal AI auditing; contestability; output evidence packets; normative material screening

1 Introduction

Legal AI systems increasingly decide what law becomes visible. Case recommenders, generated research assistants and court-facing authority reports set the field of authorities from which reasons are built. The central risk is not hallucinated citations alone; it is authority selection without procedural status.

This article's claim is direct: procedural status should follow auditability, not intelligence. A fluent answer with weak sources remains reference information. A high-scoring ranked list with no counter-material pathway remains professional support. Only a reconstructable and contestable authority-screening chain can become procedural material. The missing category is *normative material screening*: outputs that filter, rank or summarize legal authorities without becoming legal sources or decision reasons.

The paper therefore asks three operational questions:

RQ1: How should AI-generated legal outputs be classified according to their procedural effects?

RQ2: What verifiability and contestability requirements are needed for each output class?

RQ3: How can these requirements be operationalized and stress-tested across implementation, public-output, model-output, source-grounding, adversarial, recoding and uncertainty settings?

The resulting framework has two dimensions. The output-effect dimension classifies what the output does in the legal process. The system-role dimension classifies how the system enters that process. The two dimensions are then linked to verifiability, contestability and accountability requirements. The model is intentionally output-first: it asks when a legal AI output may move from internal reference to procedural material, and when it must be downgraded or withdrawn.

This article contributes a concept, a model and an artifact. The concept is normative material screening: the intermediate procedural role of AI-generated authority lists, ranked case sets and legal-material summaries. The model separates output effect from system role and links both to a six-part audit vector with status thresholds, role caps and failure caps. The artifact is a provider-agnostic harness that implements the model and tests the paper's central contrast: public search outputs can be source-reconstructable but procedurally underqualified; raw model outputs can identify the right authorities yet fail for lack of procedural source binding; source-supported repairs can qualify when source anchors, locators and counter-materials are validated; adversarial repairs are rejected when source, locator, claim, link or counter-material gates fail; blocked outputs receive minimal repair frontiers; and score-uncertainty analysis identifies boundary cases rather than hiding threshold fragility.

AI-based case recommendation is the core case because it sits at the boundary between legal information retrieval and legal reasoning. Case recommendation systems are central to case-based reasoning and legal information retrieval, both longstanding areas of AI and Law research (Ashley, 1992; Rissland and Daniels, 1995; van Opijnen and Santos, 2017). They are also procedurally sensitive. In common-law precedent search, civil-law court-decision databases, administrative adjudication systems, arbitral award repositories and ODR platforms, ranking and omission can affect the legal materials available for argument. The framework is therefore jurisdiction-parameterized rather than jurisdiction-neutral: each legal system supplies its own authority hierarchy, while the verifiability problem remains shared.

2 Related Work and Gap

2.1 Legal information retrieval and case-based reasoning

Legal information retrieval and case-based reasoning both show why case recommendation cannot be evaluated as generic text ranking. Legal materials do not merely describe an external domain; statutes, precedents, administrative rules and court decisions may constitute or structure the legal domain itself. Legal retrieval must therefore account for hierarchy, temporality, authority, citations and professional use rather than reduce relevance to algorithmic similarity (Turtle, 1995; Rissland and Daniels, 1995; van Opijnen and Santos, 2017). Case-based reasoning reaches the same point from legal reasoning: cases matter through legally relevant similarities, differences, factors, values and precedential relations (Ashley, 1992; Bench-Capon and Sartor, 2003). The procedural risk is that a system may make some authorities salient and others invisible while presenting that ordering as ordinary search.

2.2 Benchmarks and legal retrieval models

Benchmarking work has substantially improved the technical evaluation of legal retrieval. COLIEE defines shared tasks for case-law retrieval, case-law entailment, statute retrieval and statutory entailment, with explicit datasets, runs and leaderboard metrics (Goebel et al., 2024). Neural retrieval systems such as BERT-PLI model paragraph-level interactions between a query case and candidate cases, addressing the length and structure of legal cases more directly than generic ad hoc retrieval models (Shao et al., 2020). Legal language and holding benchmarks such as LexGLUE and CaseHOLD extend evaluation to standardized legal-language understanding and legal holding selection (Chalkidis et al., 2022; Zheng et al., 2021). Recent retrieval-augmented legal benchmarks, including reasoning-focused legal retrieval and CLERC, move closer to professional research tasks by connecting retrieval to downstream legal analysis and citation support (Zheng et al., 2025; Hou et al., 2025). LegalBench further extends the evaluation agenda to a broader set of legal reasoning tasks for large language models (Guha et al., 2023).

These benchmarks are indispensable, but they answer a different question. COLIEE, LegalBench, CLERC and retrieval benchmarks ask whether a system retrieves, entails, cites or reasons correctly under task conditions. This article asks whether a particular output is procedurally eligible for use after it has been produced. Precision, recall, F1, mean reciprocal rank and entailment accuracy do not determine whether a ranked set may be attached to a court report, relied upon in a professional opinion, or incorporated into a decision reason. Procedural eligibility requires corpus records, authority labels, counter-authority handling, contestability and adoption logs that ordinary retrieval metrics do not measure.

2.3 Generated legal outputs and source verification

Generated legal interfaces make the procedural gap more important. A ranked list exposes a result set for inspection; a generated answer may compress selected materials into a fluent narrative and conceal omissions, conflicts or weak source support. Empirical work on legal hallucinations and commercial legal research tools shows that legal language fluency and retrieval augmentation cannot substitute for source verification (Dahl et al., 2024; Magesh et al., 2025). The audit task is therefore to distinguish source truth, retrieval relevance, authority status and procedural use: an output can be technically useful yet unfit for external procedural status, or partly opaque at the model level yet usable if the legal-material chain is reconstructable and contestable.

2.4 Auditability, explainability and contestability

The framework also draws on algorithmic accountability and AI auditing literature. Work on accountable algorithms emphasizes the need to review data, models and decision pathways rather than accept automated outputs as self-justifying (Kroll et al., 2017). AI auditing scholarship converts this concern into institutional practices for examining systems before and after deployment (Raji et al., 2020; Mökander, 2023). Legal and social-science work on explanation and contestability further shows that explanation matters because affected persons need a meaningful opportunity to understand and challenge consequential decisions (Wachter et al., 2017; Binns et al., 2018).

Existing audit and explainability accounts, however, usually address automated decision systems in general. Due-process and public-sector accountability work shows why affected persons need a practical ability to challenge consequential automated support rather than receive a generic explanation after the fact (Citron and Pasquale, 2014; Veale et al., 2018). That insight has not yet been translated into a domain-specific account of legal AI outputs that structure the visibility and ranking of legal authorities without themselves deciding a case. This article fills that gap by treating case recommendation as normative material screening and by specifying the audit artefacts needed to qualify such outputs for procedural use.

3 Case Retrieval as Normative Material Screening

This article calls such outputs *normative material screening outputs*. The category is not a softer name for retrieval. It identifies an intermediate procedural object: an output that filters, ranks or summarizes legal materials that may later support legal argument or decision-making, without itself creating legal authority or deciding rights and duties. Examples include AI-generated lists of similar cases, recommended statutes, ranked precedent sets, extracted holdings and counter-authority reports.

The distinction also blocks two weak shortcuts. Retrieval metrics such as precision, recall, F1 and mean reciprocal rank are necessary for system development, but they do not say whether a recommendation may be cited in a procedural report, relied upon in a judicial memorandum or incorporated into a decision reason. At the same time, not every legal AI output that influences legal work is a decision-support reason. Many outputs shape the preparation of legal reasoning without becoming reasons for a decision. The middle layer is where most case recommendation systems actually operate.

4 A Two-Dimensional Output Qualification Model

The proposed framework starts from a simple premise: the normative qualification of a legal AI output depends on both what the output does and how the system enters the procedure. These are separate dimensions.

4.1 Output-effect dimension

The output-effect dimension classifies the legal effect claimed for the output:

The classification uses four criteria: whether the output is exposed outside the user's internal workspace, whether a professional relies on it in advice or preparation, whether it filters the set of legal authorities available for argument, and whether an authorized decision-maker adopts it as part of a reasoned outcome. These criteria keep procedural status separate from model quality.

1. **Reference information.** The output supports internal search, summarization or drafting. It has no external procedural effect and is not presented as a basis for another person's legal position.
2. **Professional support output.** The output assists a lawyer, expert, mediator or other professional in forming advice. The professional remains responsible for verification and communication.
3. **Normative material screening output.** The output filters, ranks, summarizes or highlights legal authorities or other normative materials. It may shape the materials available for argument, response or judicial consideration.
4. **Decision-support reason.** The output is adopted, summarized or incorporated by an authorized human or institution as part of the reasons for a legal decision, settlement recommendation, administrative action or arbitral outcome.
5. **Failure consequence: no external legal effect.** This is not a substantive output class. It is the cap applied when core verifiability, contestability or accountability requirements are absent.

4.2 System-role dimension

The system-role dimension determines how far an output may travel procedurally. It is not a second label pasted onto the same output. In the executable protocol, $R(o)$ is a role cap: a back-office tool may produce reference information, a disclosed assistance tool may produce professional support, an auditable procedural tool may produce normative material screening, and an authorized decision-support tool may produce decision-support reasons only after the adoption gate is satisfied. An unaccountable external disposition tool is capped at no external legal effect. The same ranked case list may therefore be private reference, professional support, court-facing screening, or a decision reason depending on the role and adoption path.

5 A Verifiability-Based Audit Model

The framework can be expressed as an audit model. Let o be a legal AI recommendation output generated in a particular institutional setting. Its procedural status $PS(o)$ is a function of output effect, system role and an audit vector:

$$PS(o) = g(E(o), R(o), \mathcal{A}(o)).$$

Table 1: Output status and procedural role

Output class	Typical use	Minimum procedural question	Audit artefact required
Reference information	Internal search, summary, first-pass drafting	Can the user verify the source before relying on it externally?	Source record and user-visible citation or document link.
Professional support output	Lawyer advice, expert preparation, mediator support	Can the professional explain the basis and limitations of the output to the affected person?	Source, query, output and limitation record.
Normative material screening output	Case recommendation, authority ranking, statute or precedent set selection	Can parties or reviewers reconstruct the corpus, version, ranking logic and omitted counter-materials?	Corpus manifest, ranking record, counter-material record and review log.
Decision-support reason	Output adopted in a judgment, administrative decision, arbitral award or ODR recommendation	Can the adoption path, human reasoning and contestation record be audited?	Adoption log, human reasons, contestation record and final responsibility node.
No external legal effect	Unaccountable or unverifiable output affecting rights, duties or procedural opportunities	Is any core requirement missing so that use must be blocked or downgraded?	Failure record stating the missing gate.

Here $E(o)$ denotes output effect, $R(o)$ denotes system role, and $\mathcal{A}(o)$ denotes a six-part audit vector:

$$\mathcal{A}(o) = (S, Q, H, K, T, L).$$

The six components are source and corpus verifiability (S), query and version traceability (Q), authority-hierarchy representation (H), ranking and counter-material verifiability (K), contestability support (T), and adoption logging with audit responsibility (L). Each component is scored on a three-point ordinal scale: 0 means absent, 1 means partially available, and 2 means procedurally reviewable. The model is a status-allocation protocol: it determines the highest procedural status that the output may claim given the audit evidence available.

5.1 Formal definitions and invariants

Let the procedural statuses be ordered as

$$\mathcal{P} = \{p_0, p_1, p_2, p_3, p_4\},$$

where p_0 is no external legal effect, p_1 is reference information, p_2 is professional support output, p_3 is normative material screening output and p_4 is decision-support reason. The order $p_0 \prec p_1 \prec p_2 \prec p_3 \prec p_4$ expresses the maximum procedural status an output may claim. Let $\mathcal{A}(o) \in \{0, 1, 2\}^6$ be the audit vector and let $\Pi = (\tau_N, \tau_D, \Gamma, \Delta)$ be a status policy, where τ_N is the normative-screening threshold, τ_D is the decision-support threshold, Γ is the set of necessary gate predicates and Δ is the failure-flag cap.

System role is coded as $R(o) \in \mathcal{R}$, where $\mathcal{R}$ contains back-office tools, disclosed assistance tools, auditable procedural tools, authorized decision-support tools and unaccountable external disposition tools. Define a role-cap function

$$c(R) = \begin{cases} p_1, & R = \text{back-office tool}, \\ p_2, & R = \text{disclosed assistance tool}, \\ p_3, & R = \text{auditable procedural tool}, \\ p_4, & R = \text{authorized decision-support tool}, \\ p_0, & R = \text{unaccountable external disposition tool}. \end{cases}$$

Output effect sets the claimed status; system role sets the maximum status available in the procedure. The operational rule then computes the highest procedurally available status from the audit vector, the role cap and the relevant artefact gates. Let $B_A(o)$ denote authority-set completeness, $B_E(o)$ evidence-packet completeness, $B_P(o)$ procedural source-tag completeness, $B_C(o)$ counter-material completeness, $B_J(o)$ jurisdiction-profile support, $B_T(o)$ review and contestability support, $B_D(o)$ completed adoption support and $B_F(o)$ the absence of an active downgrade, suspension or withdrawal flag. For a candidate status p, define a gate predicate $m_p(o,\Pi) \in \{0,1\}$:

$$\begin{aligned}
m_{p_1}(o) &= \mathbf{1}[S \geq 1 \wedge Q \geq 1],\\
m_{p_2}(o) &= \mathbf{1}[m_{p_1}(o) \wedge L \geq 1],\\
m_{p_3}(o) &= \mathbf{1}[S, Q, H, K, T, L \geq 1 \wedge \sum \mathcal{A}(o) \geq \tau_N\\
&\qquad \wedge B_A(o) \wedge B_E(o) \wedge B_P(o) \wedge B_C(o) \wedge B_J(o) \wedge B_T(o) \wedge B_F(o)],\\
m_{p_4}(o) &= \mathbf{1}[m_{p_3}(o) \wedge \sum \mathcal{A}(o) \geq \tau_D \wedge T = L = 2 \wedge B_D(o)].
\end{aligned}$$

The preliminary status is the greatest status whose audit gate is satisfied:

$$\sigma_\Pi(o) = \max_{\preceq}\{p \in \mathcal{P} : m_p(o,\Pi) = 1\}.$$

Let $G_D = 1$ only when the review gate records completed review, authorized adoption, human authorization, jurisdiction assumptions, adoption reasons and contestation record. Failure flags then apply a cap operator $\kappa_F : \mathcal{P} \to \mathcal{P}$:

$$PS_\Pi(o) = \min_{\preceq}\{c(R(o)), \kappa_{F(o)}(\sigma_\Pi(o))\}.$$

Warning flags leave the status unchanged while recording a defect. Downgrade and suspension flags cap external status at reference information unless the missing artefact is restored. Withdrawal flags cap status at p_0.

The protocol has three invariants. First, *gated monotonicity*: if $\mathcal{A}' \geq \mathcal{A}$ componentwise and R, G_D and F are unchanged, then $\sigma_\Pi(\mathcal{A}') \succeq \sigma_\Pi(\mathcal{A})$ only inside the same satisfied gate. Extra score cannot compensate for a missing source, counter-material or adoption gate. Second, *cap dominance*: for any output o, $PS_\Pi(o) \preceq \sigma_\Pi(o)$ and $PS_\Pi(o) \preceq c(R(o))$, while withdrawal flags force $PS_\Pi(o) = p_0$ regardless of upstream recall. Third, *metric non-equivalence*: retrieval metrics may inform K or issue-packet construction, but they are evidence variables, not status variables. These invariants are what make the model non-compensatory: legal fluency, high recall or strong ranking cannot purchase procedural status when a required audit artefact is absent.

Three consequences follow immediately. *Separation*: for any output with perfect upstream authority coverage but a non-procedural source tag, $PS_\Pi(o) \preceq p_1$ whenever it claims external screening status, because $B_P(o) = 0$ or the source-attribution failure cap applies; missing output-source links are treated the same way. *Screening necessity*: an output cannot reach p_3 unless each audit dimension is at least partially present, the output evidence packet is structurally present, and the authority, counter-material, jurisdiction, source-tag and contestability gates are reviewable; a high total score alone is insufficient. *Adoption necessity*: an output cannot reach p_4 unless $T = L = 2$ and $B_D(o) = 1$, even if it already qualifies as screening. These are protocol properties, not empirical assumptions; the experiments below test whether the implementation enforces them on public, model-generated, repaired and adversarial packets.

5.2 Audit dimensions

Table 2: Audit dimensions for case recommendation outputs

Dimension	Score 0	Score 1	Score 2
S: **Source and corpus**	Sources or corpus cannot be reconstructed.	Individual source links exist, but corpus scope or update status is incomplete.	Source links, corpus manifest, inclusion and exclusion rules, update date and document versions are recorded.
Q: **Query and version**	Query, filters or output version are unavailable.	Some query or version information is retained.	Query, filters, issue labels, source-collection version, workflow or interface version and output identifier are reconstructable.
H: **Authority hierarchy**	Output does not distinguish legal status.	Output marks broad categories such as binding or persuasive.	Output records jurisdiction, court level, legal force, validity, later treatment and relation to governing norms.
K: **Ranking and counter-materials**	Ranking is opaque and contrary materials are not surfaced.	Broad ranking factors or some contrary materials are shown.	Ranking/re-ranking factors, top-k set, omitted-material checks and counter-authority recall are reviewable.
T: **Contestability**	Affected persons cannot inspect or challenge the output.	A professional or internal reviewer can challenge the output.	Parties or reviewers can inspect, challenge, supplement and obtain a recorded response.
L: **Adoption and responsibility**	No adoption path or responsible actor is recorded.	Output and user action are logged.	Output ID, query, result list, adopted and rejected items, reasons, timestamp and responsible reviewer are logged.

This scoring scheme deliberately avoids requiring full model-parameter transparency in every case. A legal procedure may not need source code to decide whether a ranked case list can be contested. It does need enough information to reconstruct the legal-material chain and to test whether a relevant authority was omitted, misranked, outdated or mischaracterized.

5.3 Status thresholds

The audit vector determines the maximum status available to an output. A high aggregate score does not compensate for failure of a necessary gate. The model therefore uses both minimum gates and total-score bands.

Table 3: Status allocation rules

Candidate status	Necessary gates	Additional condition
Reference information	$S \geq 1$ and $Q \geq 1$	Output remains internal or user verifies sources before external reliance.
Professional support output	$S, Q, L \geq 1$	Professional user accepts verification responsibility and can explain limits.
Normative material screening output	$S, Q, H, K, T, L \geq 1$ and $B_A, B_E, B_P, B_C, B_J, B_T, B_F = 1$	Total score $\sum \mathcal{A}(o) \geq 9$; authority set, evidence packet, source tags, counter-materials, jurisdiction assumptions and review gate are present.
Decision-support reason	Normative-screening gates, $T = L = 2$ and $B_D = 1$	Total score $\sum \mathcal{A}(o) \geq 10$ plus completed review, human authorization, reasons and recorded contestation.
No external legal effect	Any necessary gate for the claimed status is missing	Output is blocked, downgraded or retained only as internal reference.

The mapping function g is therefore explicit rather than discretionary. First, assign the highest status whose gates are satisfied:

$$g_0(o) = \begin{cases} p_4, & m_{p_4}(o) = 1, \\ p_3, & m_{p_3}(o) = 1, \\ p_2, & m_{p_2}(o) = 1, \\ p_1, & m_{p_1}(o) = 1, \\ p_0, & \text{otherwise.} \end{cases}$$

Second, apply system-role and failure caps. A disclosed assistance tool cannot exceed professional support even if the audit vector would otherwise satisfy normative screening; an auditable procedural tool cannot become a decision-support reason without authorized adoption. Withdrawal-level failures, such as unreconstructable sources, materially false generated summaries or irreversible external action without human authorization, set $PS(o)$ to no external legal effect. Suspension or downgrade flags, such as high-authority omission, invalid-authority recommendation, counter-material suppression, missing source-attribution tags, absent jurisdiction assumptions, review-gate failure, ranking drift or contestation failure, cap the output at reference status until the missing artefact is restored. Thus g combines the gate function g_0 with role and failure caps. If $K = 0$, the output cannot be treated as normative material screening because the ranking and counter-material treatment cannot be reviewed. If $T < 2$, $L < 2$ or $G_D(o) = 0$, the output cannot be a decision-support reason, even if retrieval accuracy is high. If the system purports to trigger an external disposition without legal authorization, review, contestability or accountability, the output has no external legal effect.

Operationally, an evaluator applies the model in six steps: identify the candidate procedural status claimed for the output; freeze the relevant issue, query, source collection, workflow version and output identifier; score each component of $\mathcal{A}(o)$ using the audit artefacts in Table 2; apply the necessary gates in Table 3; assign the highest available status or record the missing gate; and repeat the audit after material corpus, workflow or interface changes. This makes the framework usable as a deployment checklist, post-use audit and benchmark extension.

The thresholds use non-substitutable gates. The 0–2 scale is ordinal: 0 means absent, 1 means incomplete or internal, and 2 means reviewable by the relevant professional or procedural actor. The default $\tau_N = 9$ permits one limited weakness while still requiring every gate; decision-support status requires $T = L = 2$ because external reasons require both contestation and adoption records. The full validation suite re-evaluates all 246 packets under thresholds 8–12. Thresholds 8–10 produced identical status allocations; only stricter thresholds generated demotions, and none generated promotions.

Table 4: Full-packet sensitivity of status allocation to threshold policy

τ_N	τ_D	Flips	Demotions	Status distribution
8	10	0	0	decision 12; screening 42; professional 23; reference 137; no effect 32
9	10	0	0	decision 12; screening 42; professional 23; reference 137; no effect 32
10	11	0	0	decision 12; screening 42; professional 23; reference 137; no effect 32
11	12	2	2	decision 12; screening 40; professional 25; reference 137; no effect 32
12	13	49	49	decision 0; screening 17; professional 60; reference 137; no effect 32

5.4 Jurisdictional parameterization

The audit dimensions are stable, but the materials satisfying H and K are local. For each setting j, evaluators instantiate authority hierarchy as H_j and counter-material handling as K_j before scoring. A high-status output must also carry jurisdiction assumptions that support the declared profile; an otherwise complete packet is downgraded when the profile is absent or mismatched.

Table 5: Jurisdictional profiles for H_j and K_j

Setting	H_j: authority hierarchy	K_j: counter-materials and limiting materials
Common law	Binding precedent, persuasive precedent, statutes, subsequent treatment, overruled or superseded authorities.	Distinguishing cases, contrary lines, limiting subsequent treatment, minority lines and overruled authorities.
Civil law	Constitutional norms, code provisions, special statutes, regulations, high-court decisions, lower-court decisions and doctrinal commentary.	Competing statutory interpretations, systematic conflicts, teleological limitations, contrary doctrinal views and later amendments.
Admin. adjudication	Statutes, binding regulations, agency rules, official guidance, adjudicative decisions and non-binding policy documents.	Conflicting agency interpretations, withdrawn guidance, higher-level statutory constraints and contrary adjudicative decisions.
Arbitration	Contract text, chosen law, mandatory law, institutional rules, prior awards and trade usage.	Contrary awards, mandatory-law limits, distinguishing clauses, procedural fairness objections and public-policy exceptions.
ODR	Platform rules, consumer statutes, terms of service, regulatory guidance and prior platform decisions.	Consumer-protection overrides, exception policies, appeal decisions and human-review overrides.

5.5 Contestability and audit responsibility

Contestability requires a practical opportunity to inspect selected materials, identify missing or contrary materials, submit alternatives, request human review and receive a recorded response when the output is adopted. Adoption logging is distinct: it records whether, why and by whom the output was taken up after contestation was possible. Audit responsibility assigns each artefact to the actor that controls it: developers for capability and safety claims, deployers for corpus and interface records, professional users for issue framing and verification, and authorized adopters for decision uptake.

6 Operationalizing the Framework for Case Recommendation

AI-based case recommendation can be produced by many upstream architectures, but the audit framework does not depend on their internal mechanics. The legally relevant object is the output evidence packet: the source collection, versions, legal materials, output units, source locators, authority labels, counter-material links and adoption records that make the output reconstructable. Failures in these artefacts create different legal consequences. A stale source collection creates source and temporal risk. An ordering rule that privileges surface similarity may miss legally decisive distinctions. A generated summary may misstate holdings or flatten procedural posture. A user interface may display confirming cases while hiding contrary authorities.

The evaluation protocol should therefore separate upstream performance metrics from procedural checks. Precision, recall, F1, MRR and related metrics may help evaluate an upstream retrieval or recommendation component. Procedural evaluation asks whether the output has preserved the legal hierarchy of authorities, displayed counter-materials, distinguished binding and persuasive materials, recorded update status, and produced a contestable adoption trail. These are evaluation requirements for procedural status.

The procedural metrics can be stated directly. *Authority coverage* asks whether known high-authority materials for the issue appear in the output set. *Counter-authority recall* asks whether legally relevant contrary or limiting materials are surfaced. *Hierarchy accuracy* asks whether the output correctly labels binding, precedential, persuasive, overruled or superseded materials. *Version traceability* asks whether the source collection and upstream run state that produced the output can be reconstructed. *Procedural-source coverage* asks whether source links are bound to official, tool-verified or otherwise procedurally usable source tags rather than merely named. *Adoption-log completeness* asks whether later human use of the output is observable. These metrics do not replace upstream benchmarks; they determine whether a benchmarked output can be used procedurally.

6.1 Baseline testing

Baseline testing should cover both technical and legal dimensions. A case recommendation system should be tested against known authority sets, especially high-authority

Table 6: Decision matrix for legal AI output qualification

Output use	Minimum verifiability	Minimum contestability	Allowed procedural status
Internal legal search	Source and version record; user can inspect retrieved materials	Internal user review	Reference information
Professional advice support	Source, query, output and limitations recorded	Client or supervising professional can request explanation when output materially shaped advice	Professional support output
Court or tribunal case recommendation	Corpus scope, update date, authority hierarchy, ranking factors, counter-authority handling and version record	Parties can inspect, challenge and supplement recommended materials	Normative material screening output
Decision reasoning support	Full adoption path, human confirmation, contrary-material treatment and audit trail	Parties can contest use; decision-maker gives reasons for adoption or rejection	Decision-support reason
Unaccountable external disposition	Missing legal authorization, review, contestability or accountability chain	Not satisfied	No external legal effect

or binding materials. Missing a binding, precedential, designated or otherwise high-authority material is different from missing one marginally similar low-level case. The former may trigger withdrawal from procedural use even if aggregate retrieval metrics remain acceptable.

Baseline testing should also record database coverage, update intervals, jurisdictional boundaries, language coverage, case hierarchy, citation treatment and exclusion rules. A system trained or indexed on a narrow corpus should not be evaluated as if it were a general legal research system. A system that excludes unpublished decisions, local court decisions or outdated but historically relevant authorities should make that exclusion visible to the user.

6.2 Counter-material presentation

Case recommendation is procedurally dangerous when it confirms one line of reasoning without exposing plausible counter-materials. A useful legal recommendation system should therefore include counter-authority handling as a first-order evaluation requirement. The goal is to preserve the adversarial and reason-giving structure of legal reasoning by ensuring that contrary cases, limiting cases, overruled authorities and minority lines are visible when they are legally relevant.

For audit purposes, counter-authority recall can be measured against a manually curated issue set:

$$CAR = \frac{|\text{retrieved counter-authorities} \cap \text{known counter-authorities}|}{|\text{known counter-authorities}|}.$$

The denominator is not the universe of all contrary law. It is the set of contrary or limiting materials identified for a defined issue by the relevant court, professional reviewer or benchmark curator. This makes the measure imperfect but usable.

This requirement is particularly important for generated legal outputs. A ranked list may visibly display multiple authorities. A generated summary may compress them into a single narrative and thereby hide disagreement. Evaluation should test whether the output cites supporting materials accurately, distinguishes holding from dicta, distinguishes binding from persuasive authorities, and signals unresolved conflict instead of hallucinating consensus. Empirical work on legal hallucinations shows why source verification cannot be left to model fluency (Dahl et al., 2024).

6.3 Output evidence packets

The audit layer proposed here sits above any particular search, retrieval, database, generation, agent or manual review architecture. The framework therefore does not inspect upstream implementation details. It requires an *output evidence packet*: a provider-agnostic record that connects the legal propositions or recommended materials in an output to source identifiers, versions, locators and review status.

A minimally reviewable evidence packet should record the issue presented to the system, source collection identifier, source version, output identifier, output units, source identifiers linked to each unit, paragraph or section locators, source-attribution tag, authority labels, counter-material links, human review gate, human adoption record and objection record. This requirement is related to model and dataset documentation work, but the unit of documentation is a procedurally relevant legal output rather than a model card or dataset sheet (Gebru et al., 2021; Mitchell et al., 2019). The upstream system may be a keyword search engine, legal database, generated-output workflow, agent workflow or manual review process. The audit question is the same: can the legal-output chain be reconstructed, source-tagged, reviewed and contested?

Four structural packet metrics make this requirement operational. *Structural evidence fidelity* is the proportion of output-source links marked as supporting the output unit to which they are attached:

$$EF = \frac{|\text{output-source links that support the unit}|}{|\text{all output-source links}|}.$$

Evidence coverage is the proportion of legally material output units that have at least one reconstructable source locator:

$$EC = \frac{|\text{output units with source-locator support}|}{|\text{all legally material output units}|}.$$

Source-tag coverage measures output-source links carrying a verification-status tag:

$$STC = \frac{|\text{output-source links with source-attribution tags}|}{|\text{all output-source links}|}.$$

Procedural-source coverage is stricter:

$$PSC = \frac{|\text{links tagged official, tool-verified, user-verified or settled}|}{|\text{all output-source links}|}.$$

The tag may indicate whether the source was verified by a connected legal research tool, drawn from public metadata, supplied by the user, treated as settled background, or still requires pinpoint verification. These are packet-integrity metrics, not automated merits determinations: they test whether the output-source chain is structurally reviewable, while doctrinal support still requires professional or institutional review. For external normative screening, a tag that merely says "needs verification" records work still to be done. Low EF, EC, STC or PSC means that the output is not procedurally reconstructable even if an upstream system found relevant documents.

6.4 Review and adoption gates

Source reconstruction does not by itself authorize legal reliance. The gate record should state whether professional review is required and completed, what jurisdiction assumptions were used, which reliance level applies, whether any irreversible external action has human authorization, and what the human actor did with the output. An accurate draft remains below decision-support status until an authorized actor reviews it, adopts it and records reasons.

The minimum adoption-log schema is: output identifier, time, user role, issue, filters, source-collection version, workflow version, displayed material set, displayed counter-materials, source-attribution tags, review-gate status, adopted item, rejected item, human reason, objection or supplement submitted, response to objection and final responsibility node. A system that cannot export these fields should remain below decision-support status.

7 Downgrade and Withdrawal Conditions

Procedural status is dynamic. Corpus changes, workflow updates, ranking drift, source-attribution gaps, review-gate failures or contestation failures can trigger failure caps after an output or system was initially approved. Table 5 operationalizes those caps: warnings correct isolated defects, downgrades cap status until a missing artefact is restored, suspensions block external use for an issue class, and withdrawals remove external use when the legal-material chain cannot be reconstructed or an irreversible action lacks human authorization.

8 Implementation and Evaluation

The rules above are implemented as an executable audit harness and proof-carrying allocation layer. The implementation uses local JSON scenario files as the unit of evidence and produces Markdown, JSON and certificate artefacts. The companion artifact supplies the harness, scenario files, source snapshots, run artefacts and replication commands.

The harness implements the mapping in Section 4 directly. Each scenario declares a claimed status, system role or role-inference evidence, a base audit vector S, Q, H, K, T, L, a jurisdiction profile, upstream-performance metrics and expected outcomes. Authority sets, evidence packets, review gates and counter-material declarations are mandatory for outputs that claim or qualify for screening or decision-support

Table 7: Downgrade severity levels

Severity	Trigger	Procedural consequence
Warning	Isolated error with reconstructable source, query and adoption record.	Correct the output and monitor the issue class.
Downgrade	Missing authority hierarchy, incomplete counter-material display, missing source tags, incomplete review gate or weak adoption logging.	Limit output to reference or professional support status until the missing artefact is restored.
Suspension	Repeated omission of high-authority materials, invalid authority recommendation or unexplained ranking drift in an issue class.	Suspend external procedural use for that domain or data source.
Withdrawal	Source, corpus, ranking or adoption chain cannot be reconstructed, the system generates materially false legal summaries, or an irreversible action lacks human authorization.	Withdraw from external legal use; retain only internal testing or research use.

status; missing fields trigger caps rather than silent upgrades. The runner does not read the expected outcome when computing status. It computes the allowed status, role cap, counter-authority recall, authority coverage, invalid-authority rate, structural evidence fidelity, evidence coverage, source-tag coverage, procedural-source-tag coverage and downgrade, suspension or withdrawal disposition. Separate commands vary status thresholds and re-score the same packets under strict and lenient coding policies.

The complete evaluation matrix tests one rule: procedural status follows the legal-material chain. It contains 246 scenario packets with 679 embedded records or outputs, plus 30 public source-text anchor checks, 50 raw model-output transcript locator checks, 51,643 formal invariant checks, 201 metric-separation evaluations, 2,000 metric bootstrap/permutation resamples, 2,772 baseline-rule predictions, 336 gate-ablation evaluations, 1,674 source-chain attack variants, 270 contestation challenge variants, 1,134 metamorphic policy tests, 30 query-perturbation variants across 5 issue groups, 315 query portfolios plus 5 group frontier summaries across the same public retrieval variants, 4,474 repair-frontier counterfactuals, 395 jurisdiction-profile checks, 884 rank-window visibility checks, 76 rank-order visibility counterfactuals, 5,412 status-certificate replay checks, 3,936 certificate proof obligations, 5,415 certificate tamper-resistance cases and 4,182 policy-constants replay checks. Public-output and holdout layers test reconstruction; model-output layers test source binding and contestability interventions; metric, baseline and invariant layers test non-equivalence between upstream performance and procedural status; and the ablation, attack, contestation, certificate, replay, metamorphic, query, repair, profile and ranking layers test whether any missing link breaks qualification.

Derived robustness checks then vary thresholds, recode audit vectors under strict and lenient policies and run score-blinded codebook coding. These checks test whether the same packet evidence leads to stable status allocation across coding and threshold perturbations.

The inference target is procedural allocation. A positive packet is one that the full protocol qualifies for normative screening or decision support after applying score gates, evidence-packet gates, role caps and failure caps. Baseline-rule, metric-separation and gate-ablation layers ask whether simpler rules or missing-gate variants reproduce that reference allocation. Public-output, holdout, model-output, source-anchor and adversarial layers test how the reference allocation behaves on varied artefacts.

Across ten stress-test scenarios, all rule outcomes were reproduced. The mean audit score was 8.30 and mean upstream recall was 0.84. Three scenarios had upstream recall above 0.80 but were procedurally blocked or downgraded. A high-recall output may still be unfit for external legal use when it omits a high-authority material, suppresses a

Table 8: Claim-to-evidence map

Claim	Evidence layer	What would falsify the claim
The protocol is executable	Stress scenarios, issue ablations and unit tests	Gates, caps or downgrade rules produce the wrong status.
Public outputs are auditable without model access	Metadata, public-system outputs and endpoint-matched retrieval	Visible records cannot be frozen, source-tagged or reconstructed.
Retrieval success is not procedural qualification	Public retrieval and raw GPT-5.5 outputs	High authority coverage alone upgrades the output to screening.
Retrieval metrics are not procedural status	Metric separation analysis	Precision, recall or F1 thresholds predict screening qualification with high precision.
Source-supported repair can upgrade an output	Repaired model outputs, evidence ladder and mixed-authority issue packets	Validated source, locator, counter-material and tag evidence still fails to qualify.
The validator rejects brittle repairs	Sixty adversarial repair variants	Locator, source, claim, link or counter-material attacks pass as screening.
Manifest support is externally anchorable	Public source-text anchor verification	Manifest support terms cannot be located in public source snapshots.
Raw model-output coding is transcript-anchored	Model-output transcript verification	Scenario locators cannot be found in the committed raw transcript.
Formal status properties are executable	Formal invariant verification	Scores alone can buy screening status without authority-set, evidence-packet, counter-material, review-gate, role-cap or adoption conditions.
Qualified outputs depend on their gates	Gate ablation analysis	Removing procedural gates from qualified packets leaves status unchanged.
Blocked outputs have repair frontiers	Repair-frontier analysis	Blocked high-status claims cannot be mapped to minimal source, authority, contestability, role or adoption repairs.
Rank salience is independently status-relevant	Ranking-visibility diagnostics	Coverage-preserving rank-order drift leaves high procedural status unchanged.
Query formulation is audit-relevant	Query-perturbation stability	Issue-equivalent query variants never change authority coverage, record sets or top-result identity.
Query expansion is not a substitute for counter-material protocol	Query-portfolio frontier	Multi-query public retrieval portfolios recover high authority and counter-material sufficiently for screening.
Status allocation is replayable	Status-certificate replay	Scenario identity, scenario hash, policy body, jurisdiction profile, gates, metrics, claim support, proof obligations or derivation hash cannot be reproduced.
Status allocation is tamper-resistant	Certificate tamper-resistance	A changed certificate field, proof obligation or certificate-set membership survives replay.
Status allocation is separately replayable	Policy-constants replay	A second implementation parameterized by JSON policy constants disagrees with the emitted status certificates.

Table 9: Full evaluation matrix

Layer	Units	Rule pass	What it checks
Stress scenarios	10	10/10	Status gates, failure caps, downgrade and withdrawal behavior.
Public metadata	120 records in 6 files	6/6	Source reconstruction across six public legal-record sources.
Public-system outputs	60 records in 6 files	6/6	Real upstream legal-output reconstruction and professional-support caps.
Issue-specific public output/source packets	19 records in 3 files	3/3	Public issue-search outputs and one source-bound public-source packet tested against high-authority and counter-material requirements.
Endpoint-matched public retrieval benchmark	169 records in 22 files	22/22	Public case-law or known-item outputs tested against authority sets that the endpoint can return; mixed authority gaps are recorded separately.
Out-of-sample holdout validation	126 items in 24 files	24/24	Withheld public retrieval packets, raw model-output packets and source-bound repairs tested after the scoring policy was frozen.
Raw model outputs	10	10/10	High authority coverage without source binding remains procedurally capped.
Source-supported model-output repairs	10	10/10	Same model outputs qualify only after a validator checks manifest membership, locators, hashed source-support excerpts, source tags and issue-set coverage.
Model-output evidence ladder	70	70/70	Same model outputs move across reference, professional-support, screening, decision-support and no-effect status only when evidence and adoption gates change.
Adversarial source-support repairs	60	60/60	Negative controls test locator, claim, contradiction, source, link and counter-material attacks against the repair validator.
Public source-text anchors	30 checks	30/30	Support terms are checked against extracted public source text snapshots, reducing manifest-only validation.
Model-output transcript anchors	50 checks	50/50	Raw model-output scenario locators are checked against the committed transcript sections.
Formal invariant verification	51,643 checks	51,643/51,643	Exhaustive generated states test monotonicity, authority-set necessity, evidence-packet necessity, counter-material necessity, adoption gates, role caps, failure caps and metric non-equivalence.

Table 10: Full evaluation matrix: diagnostics and robustness

Layer	Units	Rule pass	What it checks
Metric separation	201 evaluations	Recall-threshold precision 0.27; reference gate FP 0	Tests whether upstream precision, recall and F1 reproduce protocol-defined procedural qualification.
Metric statistical robustness	2,000 resamples	Bootstrap CI for recall-threshold precision 0.21–0.33; permutation $p = 0.434$	Tests whether the retrieval/status separation is stable under resampling and label permutation.
Baseline rule comparison	2,772 predictions	Best simplified false positives 38; reference rule false positives 0	Tests whether recall, F1, total-score, source-bound or review-gate substitutes can reproduce the full audit allocation.
Qualified-output gate ablations	336 ablations	336/336	Removes procedural gates from all qualified packets and checks that status falls below the corresponding level.
Source-chain attacks	1,674 variants	1,674/1,674	Applies every nonempty combination of five source-chain attacks across qualified packets; all variants lose high status through downgrade, suspension or withdrawal despite high audit scores and high upstream recall.
Dynamic contestation challenges	270 variants	270/270	Valid challenges block high status in 216/216 cases; unsupported challenge controls preserve high status in 54/54 cases.
Metamorphic policy tests	1,134 transformations	1,134/1,134	Claim escalation, metric inflation, role-cap demotion, source-tag mutation, review-gate removal, score-and-role inflation without adoption, and benign-source augmentation satisfy the expected status-function relations.
Query-perturbation stability	30 variants / 5 groups	5/5 status-stable groups	Compares issue-equivalent public-search queries; authority coverage is unstable in 3/5 groups, returned-record sets in 4/5 groups and top results in 4/5 groups.
Query-portfolio frontier	315 portfolios / 5 groups	0/315 qualified	Enumerates all non-empty query portfolios; 56/315 recover full high-authority coverage, but 0/315 recover full counter-material or screening-material coverage.
Blocked-claim repair frontiers	4,474 counterfactuals	184/184	Computes minimal artifact-gate families needed to upgrade blocked normative-screening or decision-support claims.
Jurisdiction-profile mutations	395 checks	162/162	Valid generic profile assumptions preserve status, while missing or mismatched profile assumptions downgrade qualified packets.
Ranking-visibility diagnostics	884 window checks / 76 counterfactuals	76/76	Computes counter-material visibility across rank windows; rank-order counterfactuals preserve authority coverage and counter-material recall while activating ranking-drift caps.
Proof-carrying status certificates	5,412 checks / 3,936 obligations	5,412/5,412; 3,936/3,936	Replays proof-carrying status certificates for all 246 packets from scenario identity, scenario hash, policy hash, policy body, jurisdiction profile, score gate, role cap, missing gates, failure cap, metrics, claim support, proof obligations and derivation hash.
Certificate tamper-resistance	5,415 cases	5,415/5,415	Mutates certificate identities, hashes, policy bodies, scores, roles, gates, status fields, caps, metric bundles, proof obligations and certificate-set structure.
Policy-constants replay	4,182 checks	4,182/4,182	Recomputes score candidates, role caps, missing gates, failure caps, metrics and final status in a separate script parameterized by JSON policy constants without importing the harness model.
Mixed-authority public source-screening packets	5	5/5	Source-bound statute, case and source packets can qualify as normative screening.
Issue ablations	20	20/20	High-authority omissions, counter-material suppression, unverified source tags and missing adoption gates trigger caps.
Annotation robustness	492 recodings	234/246 stable	Strict and lenient recoding policies test status stability across all packets.
Annotation uncertainty	61,500 perturbations	0.933 sample stability; 0.924 qualified high-status stability	Fixed-seed score-noise perturbations locate threshold-boundary cases and test coding-noise robustness.
Score-blinded dual coding	222 packets, 2 passes	0.99 coder kappa; 0.93 min dimension kappa; 0.86 min base exact	Coding passes score non-holdout packets and compare status, inter-coder dimensions, derived gates and base calibration.
Threshold sensitivity	1230 evaluations	0 flips at thresholds 8–10	All 246 packets are re-evaluated under normative thresholds 8–12.

relevant limiting material, recommends invalid authority, lacks output-source evidence or prevents affected persons from contesting the output.

8.1 Public legal-record reconstruction

The harness was also run on public legal-record snapshots. One layer sampled 20 public metadata records from each of six legal systems. A second layer froze the top ten records visible in six public legal retrieval or listing streams. These layers test reconstruction: whether visible records can be frozen, source-tagged, linked to locators and assigned a procedural cap by the audit layer. In both layers, all six scenario files were classified as professional support outputs because public record visibility establishes source reconstructability, not issue-specific ranking, counter-material completeness or party-facing contestation.

8.2 Issue-specific public output/source audit

A stricter public layer then froze two issue-specific public search outputs and one source-bound public-source packet, converting their visible records into evidence packets. The U.S. packet used a CourtListener issue search for agency deference after *Loper Bright*; the English-law packet used a National Archives issue search for mesothelioma causation after *Fairchild*; and the EU packet used selected Legislation.gov.uk and CURIA public source pages for GDPR Article 15 access-right materials. The layer contains 19 public records across three files.

The source-bound EU public-source packet qualified as normative material screening because it included the relevant high-authority materials, limiting provisions, source locators and reviewable output-source links. The U.S. and English issue-search outputs were capped at reference information because the visible returned records omitted high-authority or counter-material records for the defined issue. The procedural-status result is sharper than a relevance score: public legal records can be source-reconstructable yet fail normative-screening qualification when the output does not preserve the issue-specific authority and counter-material chain.

8.3 Public retrieval benchmark

To make that result less dependent on isolated examples, the harness also freezes 22 public retrieval outputs and scores them against endpoint-compatible authority sets. The benchmark covers five issue families: U.S. agency deference after *Loper Bright*, English mesothelioma causation after *Fairchild*, Canadian administrative-law standard of review after *Vavilov*, German/EU right-to-be-forgotten review and EU GDPR Article 15 access rights. It uses CourtListener, the National Archives, Supreme

Court of Canada Lexum search, OpenLegalData and CURIA public search or list endpoints. The benchmark contains 169 returned records. The returned top-k records are the committed outputs exposed by the public endpoints for the selected queries. Its gold sets are matched to endpoint scope, so case-law endpoints are not penalized for failing to return statutes; mixed statute/case screening is tested in the source-bound issue packets.

All 22 outputs were capped at reference information. Mean high-authority recall was 0.23 and mean counter-material recall was 0.00. Source-reconstructable public search output is therefore not the same as normative material screening. A system can expose real cases, stable URLs and rank order while still failing the procedural requirement that high-authority and limiting materials for a defined legal issue be made visible together.

The query-perturbation diagnostic then grouped the 22 benchmark packets with eight withheld public-retrieval variants, giving 30 issue-equivalent query variants across five issue families. Procedural status was stable in all five groups, but authority coverage was unstable in 3/5 groups, returned-record sets in 4/5 groups and top results in 4/5 groups; mean pairwise record overlap was 0.39. The query-portfolio frontier then enumerated all 315 non-empty query portfolios over those variants. Query expansion improved high-authority coverage in one issue group and 56/315 portfolios recovered full high-authority coverage, but 0/315 recovered full counter-material coverage or qualified for screening. Within this frozen five-issue public-retrieval matrix, broader query formulation improved authority visibility but did not replace an explicit counter-material protocol.

8.4 Out-of-sample holdout validation

After the scoring policy was frozen, the harness generated a separate holdout layer: eight withheld public-retrieval packets, eight raw model-output packets and eight source-bound repair packets. The holdout scenario files do not store expected statuses. This layer tests whether the rule learned from the main matrix still separates three conditions that matter for the paper's claim: public legal search output, unsupported model text and source-bound model-output repair.

The result reproduced the separation. All eight public-retrieval holdout packets and all eight raw model-output holdout packets were capped at reference information, including nine packets with high upstream performance. All eight source-bound repairs qualified as normative material screening. The audit layer assigned status from visible artefacts rather than from provider identity, output fluency or retrieval recall.

8.5 Raw model-output pilot

The harness was also run on ten raw legal-authority recommendations generated by a single upstream model configuration, Codex GPT-5.5 with extra-high reasoning. The prompts covered the same three issue families as the positive-control packets: U.S. agency deference after *Loper Bright*, English mesothelioma causation after *Fairchild*, and EU GDPR Article 15 access rights. The model was not allowed to browse or use tools. This isolates the status question: whether fluent model text with strong authority coverage can itself claim normative screening status.

It cannot. In the coded raw-output packets, all ten outputs identified the expected high-authority and limiting materials in the issue families, producing mean upstream recall of 1.00 and counter-authority recall of 1.00. Yet every claimed normative-screening output was downgraded to reference information because the citations were marked as requiring verification rather than bound to official sources, corpus manifests, pinpoint locators and a contestation pathway. Their source-tag coverage was 1.00, but procedural-source coverage was 0.00.

The repair intervention then applies the opposite condition to the same authority sets. A validator checks each output-source link against an issue manifest, locator, hashed source-support excerpt, contradiction pattern, procedural source tag and issue-set coverage before assigning repair scores; when a limiting material is already retrieved but outside the review window, a separate rank-salience repair moves that material into view. A transcript verifier first checks that all 50 scenario locators appear in the ten committed raw-output sections. A separate public source-text layer checks the manifest support terms against extracted source snapshots; all 30 support items were externally anchored across 30 records with text snapshots. All ten repaired variants qualified as normative material screening and none became decision-support reasons. This paired result is the paper's central rule in executable form: authority coverage is not procedural status, but validator-backed source support and visible counter-material can make authority coverage procedurally screenable when contestability and logging gates are satisfied.

The evidence-ladder intervention then varies seven conditions over ten model-output variants: raw unverified output, source-bound output without counter-material, source-bound output without contestability, source-bound output without logging, contestable screening, authorized decision-support adoption and unauthorized external action. The ladder produced the predicted status pattern: 30 reference-information outputs, 10 professional-support outputs, 10 normative-screening outputs, 10 decision-support reasons and 10 outputs with no external legal effect. This is the controlled result: generated legal content changes procedural status only when the legal-material chain, rank salience or adoption gate changes.

The same ten outputs were also converted into sixty adversarial repair variants. The attacks introduced locator mismatches, unsupported claims, contradiction-pattern claims, out-of-manifest sources, missing output-source links and counter-material omissions. All sixty adversarial variants were rejected despite mean upstream recall of 1.00: forty were capped at reference information, while twenty unsupported or contradictory source-support variants were withdrawn from external legal effect. The blocked-failure distribution across the full matrix shows why high-performing outputs failed: source-attribution gaps dominated, followed by summary distortion, counter-material suppression, authority omission and smaller numbers of contestability and invalid-authority failures.

Table 11: Raw model-output pilot

Issue family	N	Mean score	Upstream recall	PSC	Allowed status
U.S. agency deference after *Loper Bright*	3	9	1.00	0.00	Reference information
English mesothelioma causation after *Fairchild*	3	9	1.00	0.00	Reference information
EU GDPR Article 15 access rights	4	9	1.00	0.00	Reference information

To test mixed-authority normative material screening itself, the harness includes five public source-screening packets: U.S. administrative law after *Loper Bright*, English-law mesothelioma causation after *Fairchild*, EU GDPR Article 15 access-right interpretation, Canadian administrative-law standard of review after *Vavilov*, and German/EU fundamental-rights review after the *Right to be Forgotten* decisions. Each packet defines high-authority materials, limiting or counter-materials, invalidated or historical treatments, source tags and output-to-source links at the level of individual output units. These mixed statute/case/source construct tests ask whether a source-bound, issue-defined authority packet can obtain screening status under the protocol. All five qualified as normative material screening, while decision-support status remained reserved for authorized institutional adoption.

The ablation suite then removes one procedural condition at a time from those packets. Twenty ablations were generated: high-authority omission, counter-material suppression, unverified source tags and missing authorized-adoption gates for each issue family. Fifteen were capped at reference information through downgrade or

Table 12: Mixed-authority public source-screening packets

Issue packet	Profile	Authority set	Metrics	Status
U.S. agency deference after *Loper Bright*	Common law	H=3; C=2; invalid=1 treatment	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening
English mesothelioma causation after *Fairchild*	Common law	H=3; C=2; invalid=1 treatment	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening
EU GDPR Article 15 access rights	Civil law/EU	H=3; C=2; invalid=2 treatments	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening
Canada standard of review after *Vavilov*	Common law	H=3; C=2; historical=1 treatment	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening
Germany/EU right-to-be-forgotten review	Civil law/EU	H=3; C=2; cross-system=2 treatments	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening

suspension. The remaining five claimed decision-support status but were capped at normative screening because the packet lacked authorized institutional adoption. This is the crucial negative test: a packet that is otherwise strong on source links and authority coverage still loses status when the missing condition is procedural rather than semantic.

The metric-separation layer uses the 201 packets with complete upstream precision, recall and F1 scores to test whether conventional retrieval metrics reproduce procedural qualification. They do not. The point-biserial correlation between upstream recall and protocol-defined qualification was 0.06. A recall threshold of 0.8 identified every procedurally qualified packet but also produced 144 reproduction false positives, yielding precision 0.27 against the reference allocation. The statistical robustness layer then ran 1,000 bootstrap resamples and 1,000 label permutations. The 95% bootstrap interval for recall-threshold precision was 0.21–0.33; the interval for high-recall blocked rate was 0.67–0.79; and the recall/status permutation test returned $p = 0.434$. A broader baseline-comparison layer then evaluated 12 alternative status rules over 2,772 predictions: recall, F1, precision, total score, score candidate, source binding, review readiness and source-plus-score combinations. Every simplified substitute failed to reproduce the full protocol. The best simplified rule–source-bound score candidate with review gate–still admitted 38 outputs that the full audit rule capped below normative screening. The result is the empirical counterpart to the formal metric non-equivalence invariant: retrieval success, high total score, source binding and human-review posture may all be necessary in some settings, but none is a status-conferring condition by itself.

The gate-ablation layer then tests gate necessity across the outputs that did qualify. For each of the 54 packets that reached normative screening or decision-support status, the harness generated counterfactual variants removing the evidence packet, replacing procedural source tags with verification-needed tags, omitting high-authority materials, omitting counter-materials, removing the review gate or imposing a professional-support role cap. For the 12 decision-support packets, it also removed the decision-adoption reason. All 336 ablations fell below the corresponding procedural level. The result matters because it shows that qualification is not carried by an isolated issue packet or a high total score; it depends on the procedural chain as a whole.

The source-chain attack layer applies the same claim at data level. For each of the 54 qualified packets, the harness generated every nonempty combination of five attacks: locator mismatch, output-source-link mismatch, nonprocedural source tags, high-authority omission and counter-material omission. All 1,674 attacked variants lost high status despite mean audit score 11.28 and mean upstream recall 0.99: 54 were downgraded, 324 were suspended and 1,296 were withdrawn; 378 were capped at reference information and 1,296 were withdrawn from external legal effect. A high-scoring output therefore cannot retain procedural status when the reconstructable legal-material chain is falsified.

The contestation-challenge layer makes the T dimension dynamic rather than merely declarative. For the same 54 qualified packets, the harness generated five challenge variants: a party-submitted counter-material challenge, a source-verification challenge, a jurisdiction-assumption challenge, a missing contestability-channel challenge and an unsupported challenge control. All 216 valid challenges blocked high status; all 54 unsupported controls preserved the original qualified status. Contestability is therefore not a ceremonial opportunity to object. It is an operational gate: a challenge matters when it breaks the source, authority, jurisdiction or review pathway, and it does not matter merely because a party disagrees.

The metamorphic policy layer tests status-function invariants without reading scenario expected labels. It applies seven transformations across the 246 primary scenario packets: claim escalation never raises allowed status; perfect upstream metrics leave status invariant; a back-office role cap blocks external status; perfect scores plus an authorized role still cannot create a decision reason without adoption artefacts; nonprocedural source tags and review-gate removal block high status; and benign procedurally tagged source additions preserve qualified status. All 1,134 metamorphic tests passed, showing that status allocation follows policy invariants rather than authored expected labels.

The repair-frontier layer applies the converse test to outputs that were blocked below their claimed high-status level. For each blocked normative-screening or decision-support claim, the harness enumerates counterfactual combinations of seven artifact-gate families: audit-vector completion, authority-packet completion, rank-salience completion, source-binding completion, review-contestability completion, role-cap completion and failure-flag resolution; decision-support claims also include adoption completion. The layer evaluated 4,474 counterfactual variants across 184 blocked high-status claims. All 184 were repairable by the declared gate families. The minimal-frontier distribution was concentrated rather than arbitrary: 108 claims required one gate family, 34 required two, 39 required three and three required four. Source-binding completion appeared most often in minimal frontiers, followed by authority-packet completion and rank-salience completion. The practical result is that downgrade is not a dead end: the protocol identifies the smallest class of missing artefact that would have to be supplied before a blocked output could be reconsidered.

The jurisdiction-profile layer tests profile-bound portability. The harness checked 233 high-status claims and found 233/233 profile checks supported. It then mutated all 54 qualified packets: replacing specific assumptions with the valid generic profile preserved status in 54/54 cases, removing the assumption downgraded 54/54 cases and substituting a mismatched profile downgraded 54/54 cases, for 162/162 profile mutations passed. The rule is portable because authority assumptions are explicit, parameterized and replayable.

The ranking-visibility layer makes salience an auditable status condition. Across 230 high-status claims, 884 rank-window visibility checks produced a counter-material visibility curve: 14/230 at rank 1, 78/223 by rank 2, top-3 counter-material visibility was 183/217, and 189/214 by rank 4. The median first counter-material rank was 3.0 and the mean reciprocal first-counter rank was 0.43. In the 76 packets with enough non-counter material to run a true rank-order intervention, the counterfactual moved counter-material below the visibility window while preserving authority coverage, counter-material recall, evidence fidelity and procedural source-tag coverage. All 76 were downgraded below normative screening, and drifted top-3 counter-material visibility was 0/76. The same authority set can be present while rank salience independently controls procedural status.

The certificate layer makes procedural status proof-carrying. For every packet, the harness records the scenario identity, scenario hash, policy hash, policy thresholds, policy body, claimed status, jurisdiction profile, score vector, score-based candidate status, role cap, missing gates, failure cap, disposition, final status, claim-support flag, metric bundle, 16 proof obligations and a derivation hash. Replaying those certificates verified 246/246 proof-carrying status certificates, 5,412/5,412 replay checks and 3,936/3,936 certificate proof obligations. The tamper-resistance layer then mutates certificate identities, scenario hashes, policy hashes, policy bodies, claimed statuses, jurisdiction profiles, score vectors, total scores, status candidates, roles, missing gates, failure flags, failure caps, dispositions, allowed statuses, claim-support flags, expected outcomes, metric bundles, proof obligations, derivation hashes and certificate-set membership. All 5,415 certificate tamper-resistance cases were rejected. A legal AI output can therefore carry its own procedural-status proof object: changing either an individual derivation field or the certificate set breaks verification.

The policy-constants replay layer is the independent executable specification. It reads `policy/legal_output_policy.json`, recomputes score candidates, inferred roles, role caps, missing gates, metrics, failure flags, dispositions, failure caps and final status without importing the harness model, and compares the result with the emitted certificates. It verified 246/246 packets and 4,182/4,182 replay checks. The status allocation must therefore be reproducible both as a proof-carrying certificate and as a second implementation parameterized by the same JSON constants.

The final layers address coding uncertainty. First, all 246 committed scenario packets were re-scored under strict and lenient policies. Status allocation was stable for 234 of 246 packets; weighted status agreement was 1.00 for base versus strict and 0.98 for base versus lenient. Second, the harness perturbed all six audit scores 61,500 times under a fixed-seed annotation-noise model. Sample-level status stability was 0.933; among the 54 qualified packets, exact-status stability was 0.883 and high-status stability was 0.924. The boundary cases were concentrated around professional-support/screening and screening/decision-support gates, where one-point coding changes affect the threshold but preserve the non-compensatory gate logic. Third, 222 non-holdout packets were converted into score-blinded packet files that removed original scores, expected outcomes, source paths and manual failure flags. Two codebook-based coding passes then applied fresh scores to the packets. Treatment names and packet facts remained visible because coders had to read the legal-output evidence. The passes reached 0.99 exact inter-coder agreement, 0.99 Cohen's kappa and 0.96 quadratic weighted kappa. At the inter-coder score-vector level, the weakest dimension was authority hierarchy (H), with minimum dimension kappa 0.93; the other five dimensions were 1.00. Derived failure-flag exact agreement was 0.97 and derived missing-gate exact agreement was 0.98. Against the base harness allocation, the comparison is status-level: the weaker coder reached 0.95 exact agreement, 0.97 weighted agreement, 0.92 kappa and 0.90 weighted kappa. The weakest base-dimension Cohen's kappa was Q=0.37, while exact agreement on that same three-level score was 0.96. Base-dimension calibration otherwise showed minimum exact agreement 0.86, minimum three-category prevalence-adjusted bias-adjusted kappa (PABAK) 0.79 and maximum mean absolute score drift 0.14. The disagreements identify the fragile gates: decision-support adoption, deliberately thin stress packets and ablations near the screening threshold.

9 Implications

The framework has three implications for legal AI evaluation. First, model performance is not procedural permission. A system may meet technical retrieval benchmarks while failing procedural qualification because the output cannot be reconstructed, source-tagged or contested. Second, human oversight is not magic. Oversight must leave review status, reasons, rejection records and adoption trails; a human signature without verifiable review should not upgrade an output's status. Third, accountability should follow control. Developers, deployers, professional users and institutional adopters control different risks and should carry corresponding explanation duties.

The framework sets the evaluation target for deployed systems: a legal AI output earns procedural status only by carrying the source chain, counter-material pathway, contestability route, adoption record and proof certificate required for that status. The validation makes this target executable: public outputs, model outputs, source-bound repairs, adversarial variants, ablations, contestation challenges, replay checks, rank counterfactuals and coding-uncertainty tests all converge on the same allocation rule.

The next empirical step is therefore clear rather than open-ended. Deployed case recommendation systems and larger legal-output benchmarks should be scored for corpus verifiability, counter-material presentation, ranking traceability, adoption logging and contestation support, and those scores should be compared with retrieval metrics and user behavior. The question for future work is not only whether legal AI systems are accurate, but whether their outputs are legally usable.

10 Conclusion

Status follows auditability, not intelligence. That is the paper's central rule. A case recommendation system may be technically sophisticated, fluent or empirically strong and still have no procedural status if its legal-material chain cannot be reconstructed and challenged. Conversely, an output does not need to be a legal source or autonomous decision-maker to matter procedurally. Once it structures the visibility and ranking of authorities, it occupies the middle category of normative material screening.

The proposed framework makes that category operational. It separates output effect from system role and links both to source, corpus, ranking, counter-material, contestability, adoption and audit requirements. It allows an output to remain internal reference information, become professional support, qualify as normative material screening, or enter a decision reason only through accountable human adoption. It also supplies downgrade and withdrawal rules when the evidentiary and procedural chain fails.

The contribution is a stricter evaluation logic for legal AI outputs: no source chain, no status; no counter-material pathway, no screening qualification; no adoption record, no decision reason.

Data and Code Availability

The companion artifact contains the audit harness, scenario packets, public snapshots, raw model-output transcript, validation scripts and generated Markdown/JSON reports. The primary replication command is `pythonscripts/run_full_validation.py`; it reruns the full validation matrix and writes the aggregate report under `experiments/full_validation/results/`. The committed public-source snapshots and raw model outputs make the reported results deterministic under local review; explicit refresh commands support fresh public collection.

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